

Introduction to Fintech

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About the Instructor

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Education: Kent State University → Northeastern University → Georgia State University (Finance)

Industry: Consulting and technical analysis in securities

Teaching: Corporate Finance, Valuation, Introduction to FinTech

Research: Corporate Governance, M&A, Blockchain, ML/NLP/LLM, Labor and Finance

Why FinTech Matters to You

Every finance career — banking, asset management, consulting, investing — will be reshaped by technology. This course gives you a head start on understanding what is changing and why.

Course Philosophy

We combine theory, real data, and hands-on analysis. You will leave able to evaluate FinTech companies critically, not just admire them.

Overview of Financial Technology

The FinTech Revolution: Mapping the Landscape

Financial services

The fintech revolution

A wave of startups is changing finance—for the better

The
Economist

May 9th 2015

From payments to wealth management, from peer-to-peer lending to crowdfunding, a new generation of startups is taking aim at the heart of the industry—and a pot of revenues that Goldman Sachs estimates is worth \$4.7 trillion. Like other disrupters from Silicon Valley, “fintech” firms are growing fast. They attracted \$12 billion of investment in 2014, up from \$4 billion the year before.

What Is FinTech? Definition and Scope

Financial Technology: The application of information technology to deliver financial services faster, cheaper, and to more people

Scope: Payments, lending, investing, insurance, compliance, banking infrastructure — every corner of finance is affected

Scale: Global FinTech investment exceeded \$250 billion from 2020–2023; over 300 FinTech unicorns exist today

Why now: Smartphones, cloud computing, AI/ML, open APIs, and post-2008 regulatory openness converged simultaneously

Key Insight

FinTech is not just an industry — it is a fundamental transformation of how money moves, is stored, lent, and invested.

THE WALL STREET JOURNAL.

Big Banks Stake Fintech Claims With Patent Application Surge

By KIM S. NASH

May 10, 2016 4:44 pm ET

Big banks and credit card companies, looking to protect their investments in innovation, have accelerated efforts to patent methods linked to key technologies. The most active areas are mobile systems, Internet of Things and data analytics. These are areas where startup financial technology firms are thriving, funded with \$21.6 billion in venture capital last year alone.

What Does FinTech Include? The Full Spectrum

Payments: mobile wallets, P2P transfers, cross-border rails (Alipay, Stripe, Wise)

Lending: marketplace lending, BNPL, AI underwriting (LendingClub, Affirm, Upstart)

Banking: neobanks, digital-only accounts, BaaS (Chime, Revolut, Nubank)

Insurance: InsurTech, telematics, AI claims (Lemonade, Root, ZhongAn)

Investing: robo-advisors, fractional shares, social trading (Betterment, Robinhood, eToro)

Blockchain & Crypto: DeFi, stablecoins, tokenization (Bitcoin, Ethereum, Uniswap)

RegTech: compliance automation, KYC/AML, fraud detection (Chainalysis, ComplyAdvantage)

Infrastructure: open banking APIs, cloud banking, data aggregation (Plaid, Marqeta, Galileo)

Five Forces Driving FinTech Today

Technology Enablers

Smartphones: 6.9 billion smartphone users globally — every person is a potential FinTech customer

Cloud Computing: AWS/Azure cut infrastructure cost by 90%, letting startups scale instantly

AI and Machine Learning: Credit scoring, fraud detection, robo-advisory at near-zero marginal cost

Structural Enablers

Open APIs: PSD2 in Europe and open banking globally force banks to share data with third-party apps

Post-2008 Regulation: New entrants faced fewer legacy rules; regulators sought to promote competition

Consumer Demand Shifts

Millennials and Gen Z: 71% of millennials would rather visit a dentist than listen to their bank (Millennial Disruption Index)

Underserved markets: 1.4 billion adults remain unbanked globally — FinTech is their first access to financial services

Key Lesson

The five forces — smartphones, cloud, AI, open APIs, and regulation — did not appear one at a time. They converged around 2010, creating a once-in-a-generation window for disruption.

What Should FinTech Bring to Us?

Top Fintech Trends 2025 to Look Forward

1. Cybersecurity and biometrics



6. Biometrics & enhanced security



2. Open banking



7. Sustainable & ESG-focused fintech



3. Regulatory technology



8. Digital-only banking & Neobanks



4. Buy now, pay later



9. Embedded Finance



5. Hyper-personalization



10. Robotic process automation



A Brief History of Financial Innovation

 **\$882.30 billion**

the projected **market value** of the industry by 2030 (Fortune Business Insights)

 **13.100**

the number of **fintech startups** in the Americas as of January 2024 (Statista)

 **\$26.67 billion**

the expected value of **AI in fintech** by 2026 (Mordor Intelligence)

 **\$10.50 trillion**

the projected total **digital payment transaction** value by 2025 (Statista)

 **\$730.5 billion**

the expected market value of **embedded finance** by 2032 (Global Market Insights)

 **\$360.4 million**

the expected value of the **personal finance software** market by 2030 (Business Research Insights)

 **110.9 million**

the expected number of bank **chatbot users** by 2026 (Insider Intelligence)

 **992.50 million**

the expected number of **crypto users** by 2028 (Statista)

 **39.1 million**

the expected number of users who will have at least one **neobank account** in 2025 (Statista)

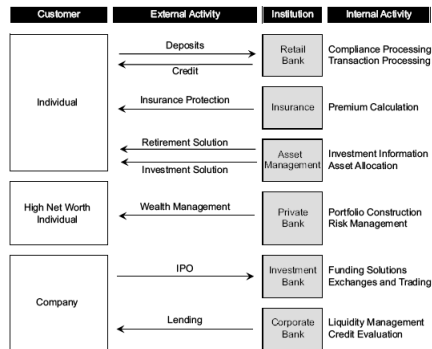
Financial Institutions and Their Activities

Deposit-taking: Commercial banks accept deposits and make loans — the foundation of money creation

Investment services: Brokers, asset managers, and investment banks channel savings into productive investments

Risk transfer: Insurance companies and derivatives markets redistribute risk across the economy

Payments: Payment networks move trillions of dollars daily with near-zero failure rates



Key Lesson

FinTech disrupts each of these functions differently. Understanding the traditional model is the first step to understanding why disruption succeeds.

Source: Gupta and Tham (2018)

Traditional Financial Institutions: The Incumbents

Payments

Visa and Mastercard: Together process over \$15 trillion in transactions annually

Business model: Charge 1.5–3% per transaction as interchange fees

Consumer Banking and Lending

Bank of America, Citibank, Chase: Earn net interest margins of 2–4% on trillions in loans

Operating cost: Maintaining branch networks costs \$5–7 per dollar lent — FinTechs operate at \$2 or less

Investment Banking and Asset Management

Goldman Sachs, Morgan Stanley: Advisory fees of 1–7% on M&A deals

Fidelity, Vanguard, BlackRock: Manage \$30+ trillion in assets; charging 0.5–2% annually

Bridgewater: World's largest hedge fund at \$150B AUM; charges 2-and-20

Key Lesson

High margins are both the incumbents' strength and their weakness — they create massive profit pools that attract FinTech disruption.

FinTech Disruption: Every Sector Under Attack

Payments and Transfers

PayPal, Venmo, Square: Processing \$2+ trillion annually; Venmo alone grew 135% YoY at its peak

Alipay: Processes \$118 trillion per year in China — more than Visa and Mastercard combined globally

Lending and Credit

LendingClub, Upstart: ML models approve loans in minutes; operating cost 60% lower than banks

Robo-advisory: Wealthfront and Betterment charge 0.25% vs. 1–2% for human advisors

Blockchain and Crypto

Bitcoin: \$1.3 trillion market cap; SEC approved spot ETFs in January 2024

Ethereum: Underpins DeFi with \$50B+ in locked value

Insurance and RegTech

Lemonade: AI-powered claims; pays some claims in 3 seconds

RegTech: KYC automation saves banks \$1B+ annually in compliance costs

Key Lesson

No segment of finance is immune. The question is not *whether* your sector will be disrupted, but *when* and *how*.

The Impact of FinTech: Businesses and Consumers

Impact on Businesses

Efficiency: Automation cuts loan processing from weeks to minutes; fraud detection improves by 40–60%

Competition: New entrants challenge incumbents, forcing fee reductions and product innovation

Acquisition: Visa, Mastercard, and JPMorgan have collectively spent \$30B+ acquiring FinTech firms since 2015

Impact on Consumers

Better service: Mobile-first banking available 24/7 with instant notifications

Lower costs: Remittances via Wise cost under 1% vs. 7–8% at traditional banks — saving migrants billions

Inclusion: 1.4 billion unbanked people gain first-ever access to savings, credit, and insurance

Key Lesson

The biggest winners from FinTech may not be tech investors — they may be the billions of people who gain their first access to affordable financial services.

Disruption: Why Incumbents Fail

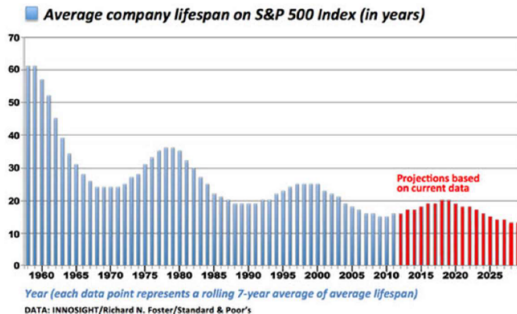
Harder to Stay on Top: The Shrinking Corporate Lifespan

S&P 500 lifespan: Average company tenure fell from 61 years (1958) to under 18 years (2012) — digital disruption is accelerating corporate turnover

Implication: At the current churn rate, 75% of the S&P 500 will be replaced by 2027

Finance is not immune: Lehman Brothers survived 158 years, then failed in 10 days; Washington Mutual collapsed in the largest U.S. bank failure ever

Agility beats scale: Smaller FinTechs with no legacy systems can deploy new features in days vs. years for incumbents



Lower Barriers to Entry: The Startup Cost Collapse

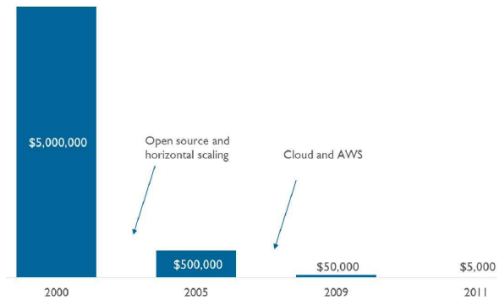
Cost collapse: Launching a tech startup fell from ~\$5 million (2000) to ~\$5,000 (2011) — a 1,000× reduction in 11 years

Cloud computing: AWS, GCP, and Azure let startups rent server capacity by the hour instead of buying hardware

Open source: Python, TensorFlow, and Hugging Face provide production-grade AI tools at zero cost

Result: More startups challenging incumbents simultaneously than at any point in history

Cost to Launch a Tech Startup



Key Lesson

When the cost of experimentation approaches zero, the number of experiments approaches infinity.

Source: Upfront Ventures

Disruptive Innovation: Three Types That Matter

Low-End Disruption

New companies start by serving overlooked, low-margin customers

Incumbents retreat upmarket, ceding low-end customers voluntarily

Example: LendingClub initially served borrowers banks rejected; now competes for prime customers

New-Market Disruption

Create entirely new customers from non-consumers

Example: Venmo created a “social payments” market that did not exist before

Example: M-Pesa gave 30 million Kenyans their first bank account via mobile phone

Big-Bang Disruption

Startups attack incumbents on *all* dimensions simultaneously: better, cheaper, and more customizable

Example: Google Maps eliminated the GPS device industry in 18 months

FinTech analog: Robinhood eliminated trading commissions industry-wide in 2019 — forced Schwab, TD, Fidelity to follow within days

Key Lesson

Christensen's disruption theory (1997) predicted low-end and new-market attacks. Big-bang disruption is a newer, faster, more dangerous pattern for incumbents.

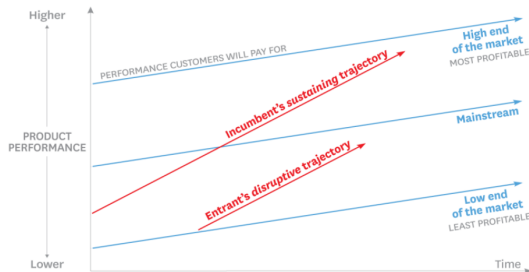
The Disruption Trajectory: How New Entrants Move Upmarket

Stage 1: New entrants offer “good enough” solutions at low cost, serving customers incumbents ignore

Stage 2: Technology improves; performance increases faster than customer needs, enabling upmarket movement

Stage 3: Disruptor competes directly with incumbents for core, high-margin customers

FinTech example: Betterment started with \$0 minimums, now manages \$45B and targets high-net-worth clients with human advisor overlays



SOURCE CLAYTON M. CHRISTENSEN, MICHAEL RAYNOR, AND RORY MCDONALD
FROM "WHAT IS DISRUPTIVE INNOVATION?" DECEMBER 2015

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Economics of Big-Bang Disruption

Three Declining Cost Curves

Creation costs: Raw materials, hardware, and software costs fall exponentially (Moore's Law)

Information costs: Social networks, reviews, and viral sharing replace expensive marketing

Experimentation costs: A/B testing on millions of users in real time vs. expensive focus groups

The “Shark Fin” Effect

Big-bang disruptors reach mass market penetration extremely rapidly — the adoption curve is nearly vertical

Incumbents have almost no time to respond before major market share is permanently lost

Example: Robinhood reached 10 million accounts in 4 years; it took Merrill Lynch 30 years to reach the same

Key Lesson

The strategy “wait and see” is fatal in big-bang disruption. By the time the threat is visible, the window to respond has already closed.

Big-Bang Market Adoption: The Vertical Rise

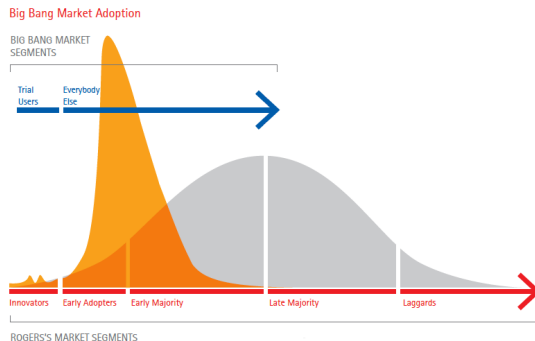
Classic S-curve: Early adopters → early majority → late majority → laggards; incumbents have years to respond

Big-bang curve: Leap directly to mass market — skipping the early-adopter chasm entirely

Why: Social networks amplify word-of-mouth instantly; free trials eliminate switching costs

FinTech example: Cash App grew from 0 to 51 million monthly active users in 6 years — faster than any prior financial product

Source: Accenture



The Shark Fin: Better, Cheaper, and More Customizable

Classic disruption: New product is cheaper but worse — incumbents can compete on quality

Big-bang disruption: New product is simultaneously better, cheaper, and more personalized — no quality defense available

FinTech example: Robinhood was free AND had a better mobile UI than full-service brokers — incumbents had to match on price AND product simultaneously

Result: A rapid rise, brief peak, then sudden collapse in the disrupted product's market share

Source: Accenture

Big Bang Disruption Upends Conventional Wisdom

CONVENTIONAL WISDOM

Focus on only one strategic "discipline" or "generic strategy" – low cost, premium product, or customer intimacy.



Strategy

BIG BANG WISDOM

Compete on all three strategic dimensions at once. Enter the market better, cheaper and customized; innovate constantly.

First target a small group of early adopters and later enter the mainstream market.



Marketing

Market to all customer segments immediately, and be ready to scale up and then exit – swiftly.

Seek innovation in lower-cost, feature-poor technologies that meet the needs of underserved customer segments.



Innovation

Launch low-cost experiments directly in the market. Combine reusable components rather than designing from scratch.

Technology Adoption is Getting Faster Every Decade

Radio: 38 years to reach 50 million users

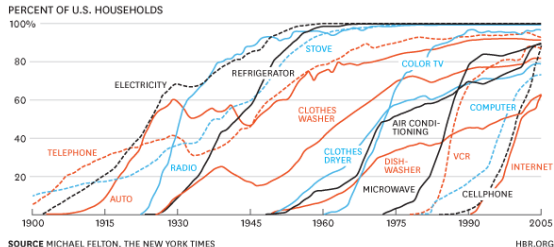
Television: 13 years to reach 50 million users

Internet: 4 years to reach 50 million users

Facebook: Under 1 year to reach 50 million users

ChatGPT: 5 days to reach 1 million users (November 2022) — the fastest technology adoption in history

CONSUMPTION SPREADS FASTER TODAY



Key Lesson

Each generation of technology is adopted 3–5× faster than the last. FinTech built on AI will be adopted faster than anything we have seen before.

The Smartphone: FinTech's Distribution Network

Adoption speed: Smartphones reached 50% of U.S. households in just 3 years — faster than any prior consumer technology

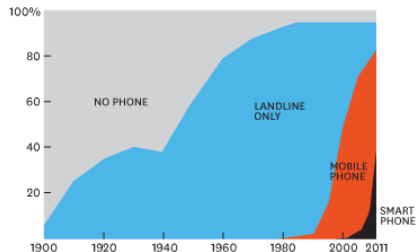
Global reach: 6.9 billion smartphone subscriptions globally; in India, smartphone users outnumber bank account holders

Distribution: Every FinTech app reaches its potential market instantly — no branches, no ATMs, no physical infrastructure

Data: Every interaction generates behavioral data for AI-based credit scoring and product personalization

FROM NO TELEPHONE TO SMART PHONES

U.S. HOUSEHOLDS BY TYPE OF PHONE, 1900-2011



SOURCE MICHAEL DEGUSTA AT THE MIT TECHNOLOGY REVIEW USING DATA FROM FORRESTER, KNOWLEDGE NETWORKS, NEW YORK TIMES, PEW, U.S. CENSUS

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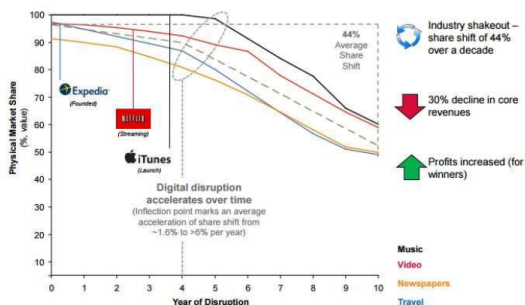
Digital Disruption is Reshaping Every Industry

Finance as prime target: High margins (bank ROE 10–15%) attract intense disruption pressure

Cross-industry: Digital disruption simultaneously reshapes retail (Amazon), media (Netflix), transportation (Uber), and healthcare

FinTech advantage: Financial products are pure information goods — no physical manufacturing required, making them ideal for digital delivery

Speed of change: McKinsey estimates 60% of current bank revenues are at risk from digitization within 10 years



Why Finance Is the Biggest Prize for Disruptors

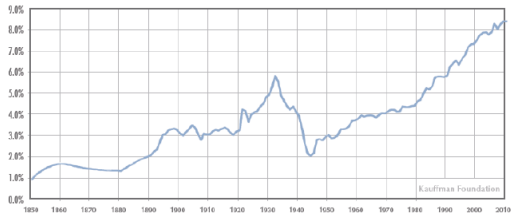
Size: Financial services represent 7–8% of U.S. GDP — approximately \$2 trillion annually

Margins: Investment banking fees of 1–7%; asset management fees of 0.5–2%; credit card interchange of 1.5–3% — all far above manufacturing margins

Inefficiency: Paper-based processes, legacy IT systems from the 1970s, and complex regulations create massive cost inefficiencies

Trust deficit: After 2008, consumer trust in banks fell sharply, making customers receptive to alternatives

Figure 1: Financial Sector as Percentage of U.S. GDP: 1850–2009



Source: The Evolution of the US Financial Industry from 1860 to 2007: Theory and Evidence. NBER.

Direct Customer Reach: Bypassing the Branch

Cost reduction: FinTechs bypass branches and brokers, reaching customers through apps — cutting distribution costs by 40–60%

Data advantage: Direct data access enables personalized offerings and alternative credit scoring beyond traditional FICO models

Scale: A single mobile app can serve 100 million customers; building 100 million branch visits is impossible

Chime example: Chime has 15 million accounts with zero branches, zero fees, and \$2.5B+ in revenue

Source: CB Insights

App Store Breakdown of Top 100 “Finance” Apps

3/29/2011 vs. 3/29/2013 vs. 3/29/2016



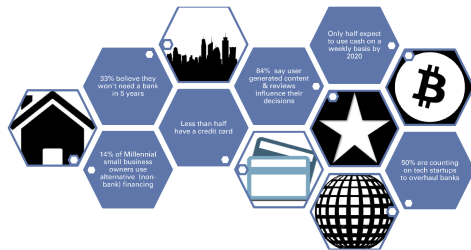
Millennials and FinTech: A Generation That Trusts Apps More Than Banks

Trust gap: 71% of millennials would rather visit a dentist than listen to their bank (Millennial Disruption Index, 2014)

Wealth transfer: \$68 trillion will be inherited by millennials from boomers by 2030 — the largest wealth transfer in history

Digital natives: Millennials and Gen Z have never known a world without the internet; paper forms and branch visits feel alien

Investment behavior: 30% of millennials hold crypto vs. under 5% of boomers — fundamentally different risk appetites



Source: Bank of America, Visa.com, Accenture, Goldman Sachs Global Investment Research

The Scale of Disruption Risk: What PwC Found

83% of financial institutions believe parts of their business are at risk from FinTechs within 5 years (PwC, 2017)

Most exposed: Payments and consumer lending face the highest near-term disruption risk

Organizational inertia: Legacy IT systems, regulatory constraints, and risk-averse culture slow incumbents' response

Response gap: Only 32% of financial institutions have a formal FinTech strategy despite 83% acknowledging the threat

82%

of incumbents expect to increase FinTech partnerships in the next three to five years

77%

expect to adopt blockchain as part of an in production system or process by 2020

20%

expected annual ROI on FinTech related projects

Source: PwC Global FinTech Report 2017

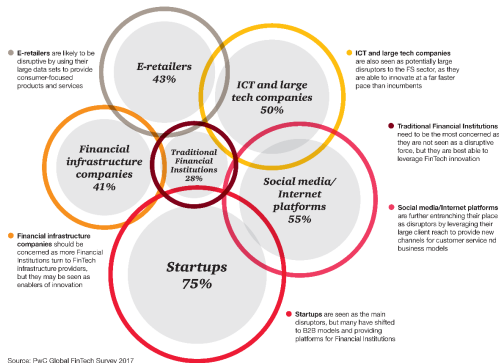
Who Are the Disruptors? Startups, BigTech, and Telecoms

FinTech startups: Move fast, specialize, and serve underserved segments (LendingClub, Robinhood, Lemonade)

BigTech: Apple Pay, Google Pay, Amazon Lending, WeChat Pay combine massive user bases, behavioral data, and brand trust — the most dangerous competitors

Telecoms: M-Pesa (Safaricom) has 30M+ users in Kenya; telecoms in Africa and Asia offer mobile money without banking licenses

Key threat: BigTech knows more about consumers than banks do — spending patterns, location, social graph



How Much Business Have Banks Already Lost to FinTech?

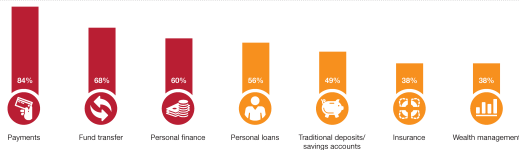
Bank perception: Banks consistently overestimate how much of their customers' business has already shifted to FinTech competitors

Reality: Actual FinTech penetration lags bank fears, but the trend line is clear and accelerating

Payment migration: PayPal, Venmo, and Zelle have captured 20–30% of person-to-person payment volume

Urgency effect: Misperception creates urgency, accelerating defensive investment even when the near-term threat is somewhat overstated

What financial activities do you believe your customers already conduct with FinTech companies?



Source: PwC Global FinTech Survey 2017

Crowdfunding and Alternative Lending

Crowdfunding: Democratizing Access to Capital

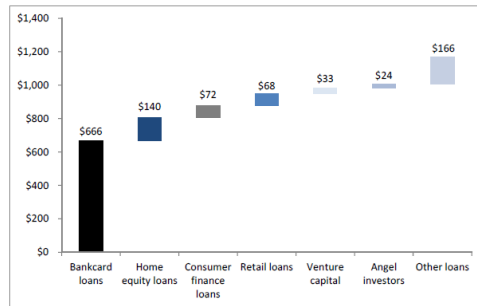
Definition: Crowdfunding platforms bypass traditional intermediaries, connecting project creators directly with investors or donors

Four types: Reward (Kickstarter), donation (GoFundMe), debt (LendingClub), equity (Republic, Wefunder)

Scale: Global crowdfunding grew from \$530M (2009) to \$34B (2015) — a 64× increase in six years

Equity threat: Equity crowdfunding directly challenges venture capital by allowing retail investors to fund startups

Exhibit 6: Addressable opportunity for crowdfunding: \$1.2 trillion
\$ in billions



Source: Federal Reserve, UNH, Goldman Sachs Global Investment Research

The Scale of Crowdfunding: \$34 Billion and Growing

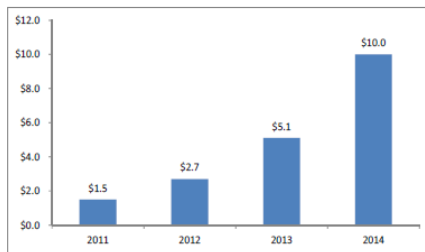
Global volume: From \$530 million (2009) to \$34 billion (2015) — 64× growth in six years

Asia-Pacific leadership: Fastest-growing region, driven by China's dominant P2P lending platforms which at their peak held \$218B in outstanding loans

Debt dominates: Debt-based lending accounts for the largest share of global crowdfunding volume

Network consolidation: Market is consolidating as network effects favor larger platforms

Exhibit 7: Aggregate amount of funding through crowdfunding
\$ in billions



Source: Crowdfund Insider

Crowdfunding by Type: Debt Leads the Way

Debt crowdfunding: Largest share of global volume, reflecting demand from underserved small businesses that banks reject

Equity crowdfunding: Fastest growing after JOBS Act (2012) legalized general solicitation

Reward crowdfunding: Kickstarter has funded 250,000+ projects, including the Pebble smartwatch (\$20M raised)

Platform consolidation: Network effects favor larger platforms; smaller platforms are being acquired or exiting

Exhibit 3: Summary of key crowdfunding platforms

	Founded	Participants (000)	Amount pledged to date (\$mn)	Projects funded	Campaign fee	Processing fee
Crowdfunding: Donation or reward based model						
Kickstarter	2009	8,092	1,573	79,571	5%	3-5%
Indiegogo	2008		170	142,301	4%	3-5% + \$25 wire
Gofundme	2010	8,000	780		5%	2.9% + \$0.30
Equity crowdfunding: Investing model						
Angellist	2010	21	200		10% carry	
Crowdfunder	2011	92	183	21,671		
Fundable	2012		154		\$179/month	3.5% + \$0.30
CircleUp	2011		50			

Source: Company data, Goldman Sachs Global Investment Research

What Drives Crowdfunding Growth?

Technology and Platform Innovation

Platform diversity: Reward, equity, debt, and donation models serve different needs and attract different participants

Viral growth: Users share campaigns through social networks, converting passive observers into active backers for free

Network effects: Each new user increases platform value for all others (Metcalfe's Law: $\text{value} \propto n^2$)

Demographic and Regulatory Shifts

Millennials and Gen Z: Comfortable with online platforms, giving online, and investing digitally

JOBS Act (2012): Reduced barriers for small companies to raise capital; Regulation A+ allows up to \$75M raised from retail investors

Global frameworks: 40+ countries have adopted similar crowdfunding regulations

Key Lesson

Technology enables viral growth; regulation creates the legal foundation. Both are necessary for crowdfunding to scale.

Generational Context: Who Is Driving FinTech Adoption?

Generational Timeline

Baby Boomers: 1946–1964 — prefer branches; hold \$73T in wealth

Generation X: 1965–1980 — hybrid; early adopters of online banking

Millennials (Gen Y): 1981–1996 — primary FinTech adopters; inherit \$68T by 2030

Generation Z: 1997–2012 — digital native; even lower bank trust than millennials

Generation Alpha: 2013–2022 — will never know a world without AI

Why Generations Matter for FinTech

Millennials and Gen Z are the primary adopters of mobile payments, robo-advisors, and crypto

Gen Z shows even lower trust in traditional banks — 33% have no bank account in some markets

By 2030, millennials will be the largest wealth cohort — their preferences will dominate product design

Key Lesson

Financial products designed for Boomers will lose market share to products designed for Millennials and Gen Z.

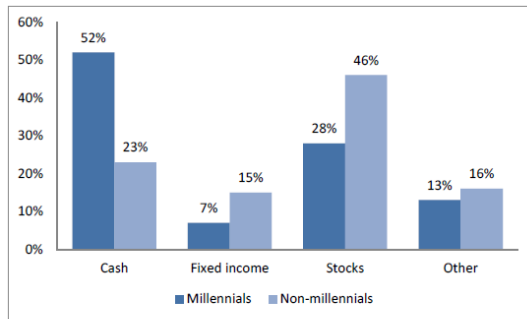
Millennials Are Less Likely to Hold Stocks

Stock ownership gap: Only 26% of Americans under 30 own stocks — the lowest in 25 years

2008 scarring: Millennials witnessed the financial crisis during formative years, reducing trust in traditional markets

FinTech opportunity: Robinhood and Acorns removed minimums and commissions — Robinhood grew from 0 to 22 million accounts in 7 years

Crypto preference: Many millennials substitute crypto for stocks — 30% hold digital assets vs. under 5% of baby boomers



Source: Gallup

Millennials Give More: The Rise of Social Giving

Giving pattern: Millennials donate more frequently but in smaller amounts, fueling donation crowdfunding platforms like GoFundMe

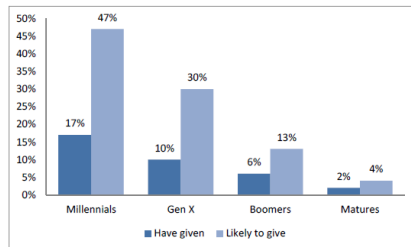
GoFundMe scale: Over \$25 billion raised across 150 million donations; 1 in 5 Americans has donated to a GoFundMe campaign

Social dimension: Campaigns spread through social networks, converting passive observers into active donors — pure viral growth

Financial inclusion: Charitable crowdfunding funds medical costs, education, and disaster relief that traditional insurance and aid miss

Exhibit 16: ...And more likely to participate in crowdfunding

% of respondents who have given or are likely to give via crowdfunding in the next 12 months



Source: Blackbaud

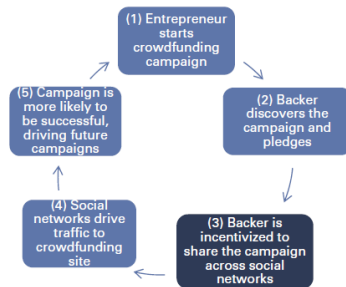
Network Effects: Why FinTech Winners Take Most of the Market

Metcalf's Law: Network value grows as n^2 — each new user makes the platform exponentially more valuable for all existing users

Payments example: Venmo is only useful if your friends use it. Once a critical mass joins, switching becomes socially costly

Winner-take-most: Alipay and WeChat Pay together hold 97%+ of Chinese mobile payment volume

Moat: Once a crowdfunding or payment platform achieves critical mass, it becomes nearly impossible to displace



Source: Goldman Sachs Global Investment Research

Regulatory Changes: The JOBS Act and Open Banking

Pre-Reform: Regulation D Barriers

General solicitation ban: Companies could not advertise securities offerings to the general public

Accredited investor requirement: Only individuals with net worth over \$1M or income over \$200K could invest in private offerings

Effect: Startup funding was restricted to a small elite; ordinary investors were locked out

Post-Reform: JOBS Act (2012)

Title II (2013): Allows general solicitation if all purchasers are verified accredited investors

Title III (2016): Regulation Crowdfunding allows non-accredited investors to invest up to \$2,200 per issuer

Title IV: Regulation A+ allows companies to raise up to \$75M from the general public

Key Lesson

Regulatory change was as important as technology in enabling crowdfunding. Policy design shapes which FinTech innovations can reach scale.

Robo-Advisory and Wealth Management

Robo-Advisory: Automated Wealth Management

Core Value Proposition

Automation: Algorithm-driven portfolio construction and rebalancing replaces human judgment with consistent, rules-based investing

Passive investing: ETFs tracking index portfolios; Vanguard showed passive beats active 80%+ of the time over 10-year periods

Ease of use: Mobile-first interface; account opening in minutes vs. days at traditional brokers

Fee and Access Revolution

Fees: 0.25% (Betterment, Wealthfront) vs. 1–2% for human advisors — saving \$750–\$1,750 per year on a \$100K portfolio

Minimums: Start at \$0–\$500 vs. \$500,000+ at traditional wealth managers

Features: Daily tax-loss harvesting, automatic rebalancing, direct indexing for large accounts

Key Lesson

Robo-advisors democratized investing: a 25-year-old with \$1,000 now receives the same portfolio science as a client with \$10 million.

Market Shares of the Largest Robo-Advisors

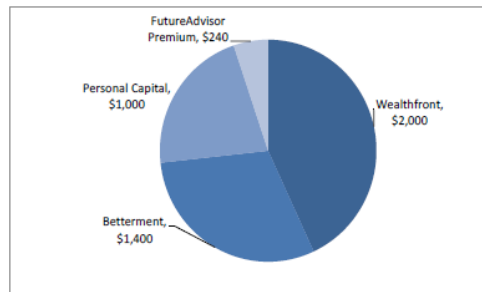
Incumbent dominance: Vanguard Digital Advisor and Schwab Intelligent Portfolios rapidly captured \$200B+ in robo-AUM by leveraging existing massive client bases

Pure-play competition: Betterment (\$45B AUM) and Wealthfront (\$30B AUM) compete on user experience, tax features, and transparency

Fees as weapon: Schwab launched at 0% management fee in 2015, forcing price compression across the entire industry

Hybrid model: Human advisors plus algorithm; JPMorgan, Goldman, and Morgan Stanley all launched hybrid services

\$ in millions, last reported AUM



Source: Company data, Goldman Sachs Global Investment Research

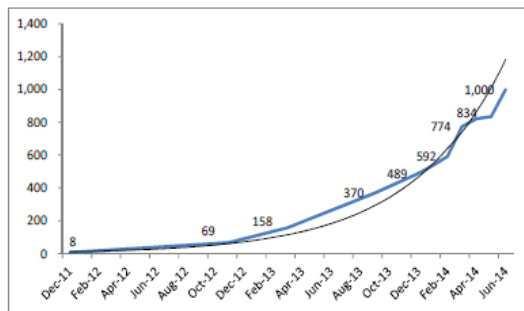
Wealthfront: From Zero to \$4 Billion in Four Years

Viral growth: Wealthfront grew from \$0 to \$4 billion in AUM within four years, driven by word-of-mouth among Silicon Valley tech workers

Target market: Young, tech-savvy professionals who were comfortable with algorithms but lacked access to wealth management

Tax-loss harvesting: Wealthfront's automated daily tax-loss harvesting generates average after-tax alpha of 1.55% annually

Industry signal: Its success validated the robo-advisor model and triggered rapid competitive response from major incumbents within 2 years



Source: Company data

Benefits of Wealthfront and Betterment: What You Actually Get

Automated tax-loss harvesting: Sells losing positions to realize losses for tax benefit, then replaces with similar ETFs — captures average \$1,500/year in tax savings on a \$100K portfolio

Daily rebalancing: Automatically maintains target asset allocation as markets move

Diversified ETF portfolios: Low-cost ETFs (average 0.07% expense ratio) across 10+ asset classes

Total cost: 0.25% fee + 0.07% ETF costs = 0.32% vs. 1–2% for comparable human-advised portfolios

Wealthfront	Betterment
• Maximize gains with passive investing • Tax aware asset allocation • No commission fees • Diversified portfolio • Lower taxes with tax loss harvesting • Hassle free investing	• Build wealth: Invest in a diversified portfolio of stock and bond ETFs designed for optimal expected returns • Save time: Everything is automated - from rebalancing to dividend reinvestment, even deposits • Save money: Our customers pay one simple all-inclusive management fee as low as 0.15% • Lower taxes: We optimize investment returns tax-efficiently, with Tax Loss Harvesting+ and other tools

Source: Wealthfront, Betterment

Charles Schwab vs. Wealthfront: Incumbent Counter-Attack

Zero-fee launch: Schwab launched Intelligent Portfolios in March 2015 with zero management fee, immediately undercutting all pure-play robo-advisors

Scale advantage: Schwab's 30 million existing clients meant instant distribution; accumulated \$10B+ in robo-AUM within 12 months

Revenue model: Schwab earns through cash allocation (2–10% cash drag) and proprietary fund fees, not advisory fees

Lesson: Decisive incumbents with scale can effectively recapture lost ground when they commit fully to a response

Source: Goldman Sachs Equity Research

Units as indicated, as of end of 2014 or most recently public information

	Schwab	Wealthfront
AUM (\$mn)	2,463,600	2,000
Employees	14,600	62
AUM/employee (\$)	168,740	32,258
Offices	300	1
AUM/office (\$mn)	8,212	2,000
Accounts (000)	9,386	222
AUM/account (\$)	262,476	9,000

Comparing Advisory Fees: The Cost Revolution

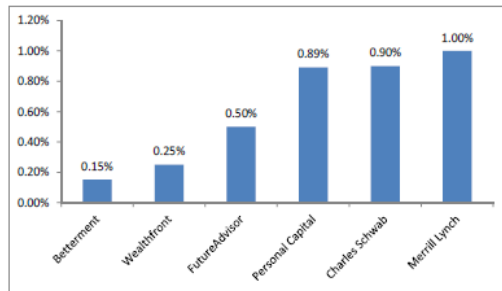
Human advisors: 1–2% per year; on a \$500K portfolio, that is \$5,000–\$10,000 annually, compounding to over \$200K in fees over 20 years

Robo-advisors: 0.25%–0.50%; same portfolio costs \$1,250–\$2,500 per year

Zero-fee model: Schwab and Fidelity at 0% management fee; revenue earned through cash holdings and proprietary funds

Democratization: Fee compression means young investors with small balances can afford portfolio management for the first time

Advisory fee as % of first \$100k managed



Source: Company data, Goldman Sachs Global Investment Research

Investment Minimums: Who Gets Access to Wealth Management?

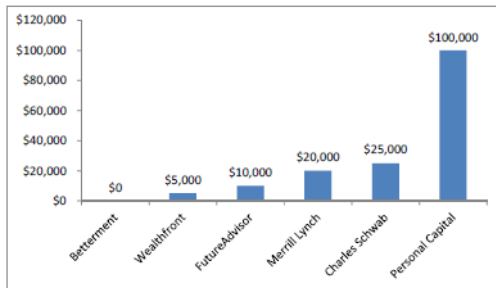
Traditional wealth management: Requires \$500,000–\$1,000,000+ in assets to access a dedicated human advisor

Robo-advisors: Start at \$0 (Betterment) to \$500 (Wealthfront) — democratizing investing for anyone with a smartphone

Micro-investing: Acorns rounds up purchases and invests the change; 10 million users invest an average of \$30/month

Millennial impact: Low minimums matter most for the 75% of millennials who have under \$50,000 in savings

Investment minimum



Source: Company data, Goldman Sachs Global Investment Research

Alternative Lending and Digital Payments

Alternative Lending: Serving Borrowers Banks Ignore

What Makes It Different

Big data underwriting: ML models use bank transaction data, e-commerce history, social signals, and 10,000+ variables vs. banks' 30–50 FICO-based variables

Automation: Online application; instant decision; funds in 24–48 hours vs. 3–6 weeks at a bank

Cost reduction: Operating cost of 2% per dollar lent vs. 5–7% at traditional banks

Market Opportunity

Addressable market: \$870 billion in U.S. personal, student, and small business loans immediately addressable

Underserved segments: Thin-file borrowers, self-employed individuals, and small businesses represent the highest growth potential

Network effects: More borrower data improves underwriting models, attracting more borrowers and lenders

Key Lesson

The 25% of small businesses rejected by banks represent a \$200B+ annual lending market for online lenders.

Marketplace Lenders: The Major Players

LendingClub: Largest U.S. marketplace lender; originated \$8.4B in loans by 2015; now a bank holding company after acquiring Radius Bank in 2021

Prosper: Pioneer P2P lender; specializes in personal loans; uses proprietary Prosper Rating system

OnDeck and Kabbage: Small business lending; Kabbage was acquired by American Express for \$850M in 2020

Upstart: AI-based consumer lending; uses 1,600+ variables; approved 43% more borrowers than traditional models with lower default rates

Loans originated in \$ millions

	Year founded	Primary market	Primary vertical	Loans originated (\$mn)	Borrowers served
LendingClub	2007	US	Personal, small business	7,596	440,000
Prosper	2006	US	Personal	2,000	250,000
OnDeck	2006	US	Small business	2,000	25,000
SoFi	2011	US	Student	1,750	15,500
Zopa	2005	UK	Personal	1,154	80,000
Funding Circle	2010	UK, US	Small business	826	7,100
Ratesetter	2010	UK, Australia	Personal	709	83,942
Kabbage	2009	US	Small business	500	100,000

Source: Company data, Goldman Sachs Global Investment Research

The \$870 Billion Addressable Market for Marketplace Lenders

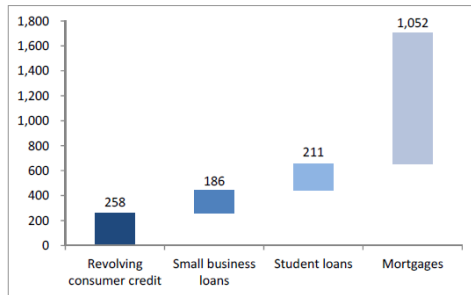
Personal loans: \$300B+ immediately addressable; high-interest credit card debt is the primary target for refinancing

Student loan refinancing: \$1.7 trillion in outstanding student debt; SoFi refinances graduates at significantly lower rates

Small business loans: \$400B+ market; 25% of applicants rejected by banks; online lenders approve in hours, not weeks

Merchant financing: Square Capital and PayPal Working Capital advance funds based on transaction history with no credit check

\$ in billions; mortgages measured as annual originations



Source: Goldman Sachs Global Investment Research

LendingClub Loan Performance: What Investors Actually Earn

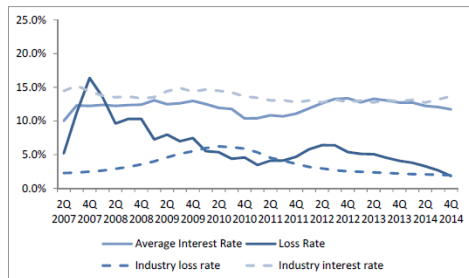
Grade A–C loans: Annualized net returns of 5–9%, outperforming savings accounts (0.5%) and money market funds (1–2%)

Diversification math: Investing \$25 across 200+ loans produces stable returns despite individual defaults; probability of losing money approaches zero with 200+ loans

Grade D–G loans: Higher yields (10–15%) but substantially higher default rates; not suitable for risk-averse investors

Institutional capital: Hedge funds and banks eventually crowded out retail investors due to faster automated buying

Average interest rate and loss rate vs. credit card interest rate and loss rate



Source: Company data

Prosper Loan Performance: Platform Comparison

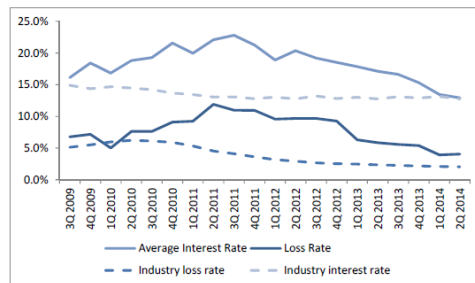
Similar returns: Prosper shows returns comparable to LendingClub, suggesting performance is driven by market conditions rather than platform-specific advantages

Credit cycle risk: Both platforms launched post-2008 and had not been tested through a full credit cycle downturn when this data was published

2020 test: COVID-19 was the first real stress test; government stimulus prevented mass defaults; true cycle testing remains incomplete

Survivor bias: Many weaker P2P platforms failed; remaining platforms represent a selected sample

Average interest rate and loss rate vs. credit card industry rate and loss rate



Source: Company data

Small Business Owners' Attitudes toward Online Lending

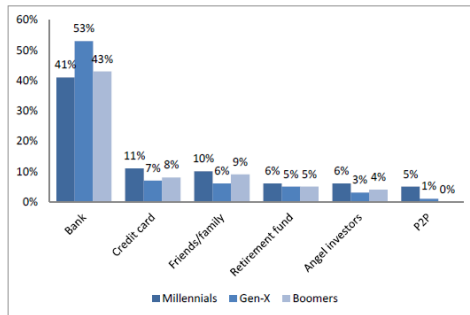
Speed advantage: Small business owners cite approval in hours vs. weeks as the primary advantage of online lenders — cash flow cannot wait 6 weeks

Rejection rate: 25% of small businesses seeking bank financing are rejected; online lenders approve many of these at higher rates (APR 10–40%)

Accessibility: No collateral requirement for many online loans; traditional banks require real estate or business assets as security

Trade-off: Speed and accessibility come at a cost — online small business loan rates are typically 2–5× higher than bank loans

% of respondents who would consider each source for SMB loans



Source: Bank of America

Operating Cost Advantage: LendingClub vs. Banks

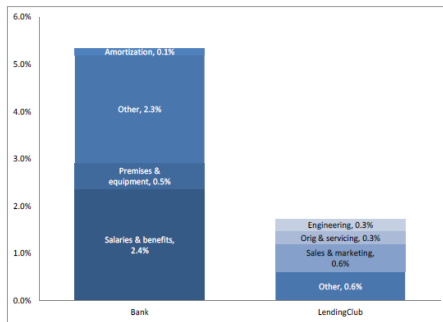
LendingClub cost: ~2% per dollar lent — automated underwriting, no branches, lower regulatory overhead

Traditional bank cost: 5–7% per dollar lent — branch network, manual underwriting, Basel III capital requirements

Cost advantage: 2.5–3.5× structural cost advantage; even with higher default rates, online lenders can be more profitable

Regulatory moat: Banks carry deposit insurance, which lowers funding costs by 1–2%; offsetting part of the operating cost disadvantage

Expenses as % of average loans outstanding



Source: Company data, FDIC, Goldman Sachs Global Investment Research

LendingClub Originations: The Fastest-Growing Loan Platform in U.S. History

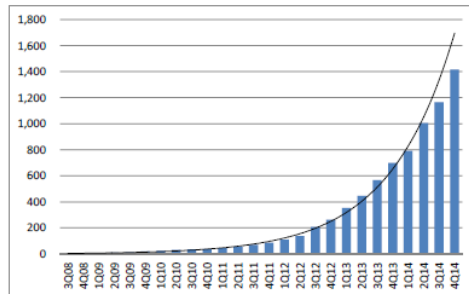
Growth trajectory: LendingClub grew from \$0 to \$8.4 billion in originations by 2015 — the fastest growth of any U.S. consumer lending platform

Platform model risk: Rapid growth requires continuously attracting both borrowers (demand side) and investors (supply side) simultaneously

2016 scandal: CEO Renaud Laplanche resigned after \$22M in loans were sold to Jefferies against its instructions, triggering a 35% stock decline

Current status: Now a bank holding company; slower but more stable funding via deposits

\$ in millions



Source: Company data, Goldman Sachs Global Investment Research

Merchant Financing: Data-Driven Lending Without Credit Checks

Square Capital: Advances funds to Square merchants based on card transaction history; repayment as a percentage of daily sales

PayPal Working Capital: Same model; has disbursed \$20B+ to small businesses since launch in 2013

No credit check: Uses transaction history as a perfect real-time indicator of business health

Natural repayment: Automatic daily repayment as a percentage of sales aligns repayment with merchant cash flow; default risk is dramatically lower

Traditional banking system	Structure	Max loan amount
C&I line of credit	Standard line of credit	Varies
Commercial card	Revolving line of credit	Varies
SBA 7(a) loans	Standard term loan	\$5,000,000
New entrants: marketplace lenders		
OnDeck	Standard term loan	\$250,000
Kabbage	1/6 of balance repaid per month	\$100,000
New entrants: merchant financing		
Amex 1 to 2 year financing	Repayment as % of receivables	\$2,000,000
Alibaba e-Credit line	Line for purchase with Alibaba supplier	\$300,000
PayPal Working Capital	Repayment as % of daily sales	8% of LTM PayPal sales
Square Capital	Repayment as % of credit card sales	--

Source: Company data, Goldman Sachs Global Investment Research

Digital Payments: The \$2 Trillion Daily Market

Scale and Growth

PayPal, Zelle, Cash App: Together processed over \$2 trillion in 2023; Venmo alone processed \$63B in 2021

Alipay dominance: Processes \$118 trillion per year in China — more than Visa and Mastercard process globally

Real-time rails: FedNow (launched 2023) enables instant U.S. bank-to-bank transfers 24/7

Why Payments Matter

Data flywheel: Every payment generates behavioral data for cross-selling loans, insurance, and investments

Remittances: \$700B+ annual market; Wise and Remitly charge under 1% vs. Western Union's 7–8%

Network effects: Payments are winner-take-most; switching costs rise with every contact that joins your network

Key Lesson

Payments are the gateway to every other financial product. Control payments and you control the customer relationship.

U.S. Mobile P2P Payments: From \$1B to \$120B in Five Years

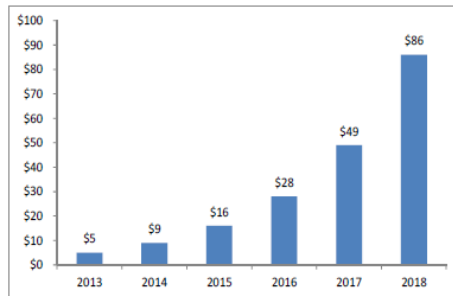
Explosive growth: U.S. mobile P2P volume grew from \$1B (2011) to \$120B+ (2016), led by Venmo, Square Cash, and Zelle

Venmo social layer: Unique social payment feed turned mundane transactions into shareable content — the core of its viral growth

Customer acquisition funnel: Users who trust a platform for splitting dinner bills adopt its credit cards, checking accounts, and loans within 18 months

Monetization: P2P transfers are free; Venmo earns 1.5% on instant transfers and 2.9% on merchant payments

\$ in billions



Source: Business Insider Intelligence

Mobile Has Become the Primary Banking Channel

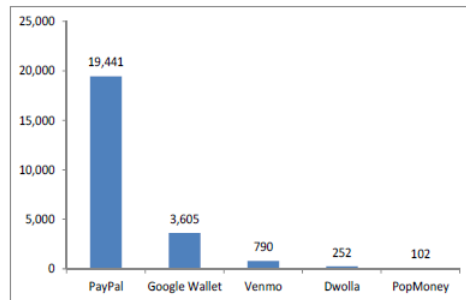
Inflection point: Mobile financial app usage surpassed in-branch bank visits in 2016 and the gap widens every year

Retention advantage: Bank customers who engage with a mobile app churn at half the rate of branch-only customers — a powerful retention moat

Engagement data: Mobile apps generate daily behavioral data; branches generate quarterly visit data at best

Cost: Serving a customer digitally costs \$0.09 per interaction; a branch interaction costs \$4.25 — a 47× cost difference

US mobile unique visitors in thousands, January 2015



Source: comScore

Electronic Payments: Huge Market, Enormous Headroom Remaining

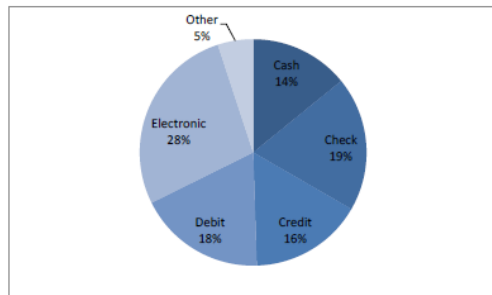
Electronic share: Electronic payments make up 28% of transaction values; cash and checks still dominate total value despite rapid digital growth

Growth rate: Electronic payments growing at 7–10% annually; cash usage declining 2–3% per year

Remaining headroom: The fact that 72% of value is still analog means the addressable market for digital payments is enormous

B2B opportunity: Business-to-business payments remain 90%+ paper-based — the next frontier for digital disruption

% of transaction value by method, 2013



Source: Federal Reserve

Venmo: Social Payments and the Viral Growth Formula

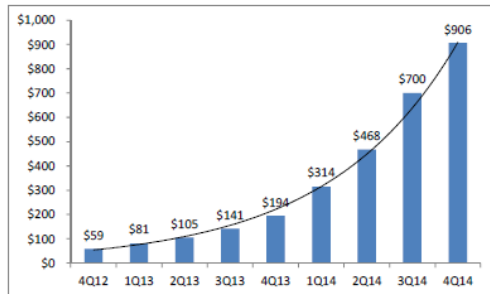
Quarterly volume: Venmo processed \$17.6 billion in Q4 2016, growing 135% year-over-year — driven by its unique social payment feed

Social layer: The public payment feed (“Venmo Moments”) creates social pressure to join and public accountability for payments — purely organic growth

Revenue model: P2P transfers are free; Venmo earns from instant transfers (1.5%), merchant payments (2.9%), and its debit card

Scale: 90 million accounts in 2023; processed \$244B in total payment volume for the year

\$ in millions



Source: Company data, Goldman Sachs Global Investment Research

The \$700 Billion Remittance Market: Ripe for Disruption

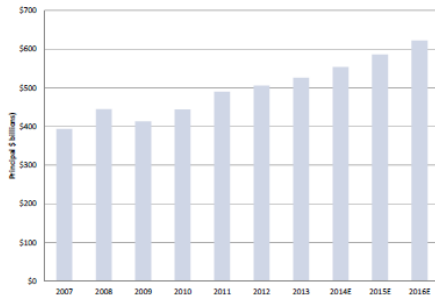
Market size: Global remittances exceeded \$601B in 2015 and topped \$800B in 2022 — larger than foreign direct investment flowing to developing nations

Economic impact: For countries like Tajikistan (51% of GDP), Lebanon (38%), and Tonga (44%), remittances are the primary source of national income

Traditional fees: Western Union and MoneyGram charge 7–8% on average; a migrant worker sending \$500/month pays \$400+ per year in fees

Digital alternative: Wise charges under 1%; Remitly under 1.5% — saving migrant workers billions annually

\$ in billions



Source: World Bank, Goldman Sachs Global Investment Research

Cost of Sending Money: FinTech vs. Traditional Banks

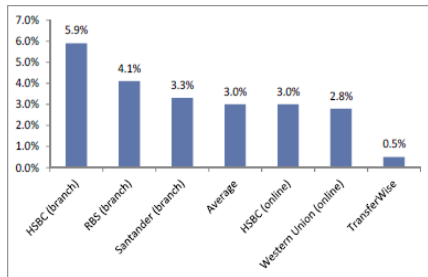
Bank cost: £20–50 to send £1,000 from UK to Germany via SWIFT — a 2–5% effective fee

Wise cost: Under £5 for the same transfer — a 4–10× cost reduction

How Wise works: Matches currency pairs locally, eliminating actual cross-border fund movements and the multi-party fee chain

Speed: Wise transfers complete in hours; SWIFT transfers take 3–5 business days and may involve 3–5 correspondent banks

Cost as % of principal



Source: Charterhouse Research

Transaction Speed: From Days to Seconds

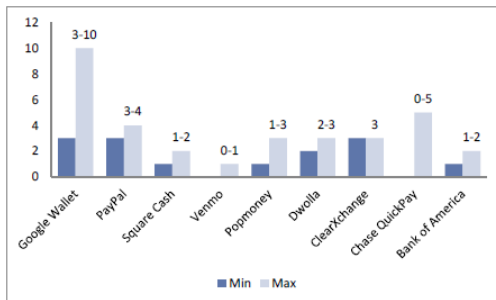
SWIFT wire: 3–5 business days; may involve multiple correspondent banks; no weekend settlement

Digital platforms (Wise, Revolut): Settlement in hours; 24/7 availability including weekends

Ripple/XRP: Settles cross-border payments in 4 seconds at under \$0.01 cost; used by Santander and other major banks for corridor payments

Business impact: Faster settlement reduces idle funds in transit and improves working capital management for businesses that wire millions daily

Business days for cash to deposit



Source: Company data, Goldman Sachs Global Investment Research

Blockchain, Crypto, and Decentralized Finance

Blockchain and Distributed Ledgers: The Trust Machine

What Blockchain Does

Decentralization: Transactions validated by a distributed network of nodes, not a central authority — no single point of failure or control

Immutability: Once recorded, transactions cannot be altered; provides an auditable, tamper-proof record

Trust without intermediaries: Smart contracts execute automatically when conditions are met — no need for lawyers, escrow agents, or clearinghouses

Applications in Finance

Payments and crypto: Bitcoin (\$1.3T cap), Ethereum, Ripple for near-instant cross-border settlement

Trade finance: HSBC and Standard Chartered use blockchain to digitize letters of credit

Securities settlement: DTCC processes \$2.2 quadrillion annually; blockchain could compress settlement from T+2 to T+0

Key Lesson

The internet transformed information flow; blockchain aims to transform value transfer — removing trusted intermediaries from every financial transaction.

Blockchain Disruption: Which Industries Are Being Transformed?

Financial Services

Payments and crypto: Bitcoin, Ethereum, Ripple — eliminating correspondent banks from cross-border settlement

Lending and crowdfunding: ICOs raised \$5B+ in 2017, bypassing investment banks to raise capital globally

Exchanges: DeFi protocols like Uniswap process \$1B+ in daily trading volume without a central exchange

Beyond Finance

Supply chains: Walmart uses blockchain for food traceability; reduces contamination investigation from 7 days to 2.2 seconds

Insurance: Parametric insurance pays automatically when smart contract conditions are met (e.g., flight delay data)

Real estate: Tokenization allows fractional ownership of \$326 trillion in global real estate assets

Key Lesson

Any industry with complex multi-party trust problems — trade finance, supply chains, insurance — is a blockchain target.

Western Union: What High Fees Cost Ordinary People

Real cost: Sending \$100 via Western Union can cost \$10–15 in combined fees and exchange-rate spread — a 10–15% effective charge on a migrant worker's earnings

World Bank target: The UN Sustainable Development Goals target reducing global remittance costs to 3%; the current global average is 6.2%

Bitcoin alternative: Bitcoin enables direct peer-to-peer transfer at under 1% cost; no Western Union account required

Viral moment: The “we fixed your ad” image went viral, crystallizing public anger at legacy fee structures

Source: Campbell Harvey

The image is a side-by-side comparison of two remittance services. The left side is for Western Union, featuring a yellow background at the top with the Western Union logo. Below the logo, the text reads "Send warm wishes today." followed by "FOR ONLY \$5/\$50" and "SEND UP TO". A small button says "Find Agent Location". The right side is for Bitcoin, featuring a grey background at the top with the Bitcoin logo. Below the logo, the text reads "Send warm satoshis today." followed by "FOR ONLY \$0.01/\$ANY" and "SEND UP TO ANY AMOUNT". A small button says "Pick Your Wallet". Both sides have a bottom section with the text "moving money for better" (Western Union) and "moving money far better" (Bitcoin). The Western Union side also includes a small image of a yellow mug and a Christmas tree, while the Bitcoin side includes a small image of a full moon and a person on a horse.

Western Union	Bitcoin
Send warm wishes today.	Send warm satoshis today.
FOR ONLY \$5/\$50	FOR ONLY \$0.01/\$ANY
SEND UP TO \$50	SEND UP TO ANY AMOUNT
Find Agent Location	Pick Your Wallet
moving money for better	moving money <u>far</u> better

Blockchain vs. the Internet: History Rhymes

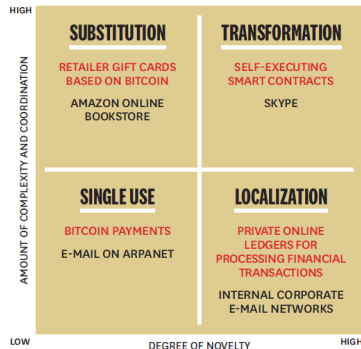
1990s parallel: The internet faced identical skepticism (“no one will put their credit card online”); early killer apps seemed trivial (email); adoption felt slow, then suddenly explosive

Foundational infrastructure: Like TCP/IP, blockchain is a foundational protocol; its transformative applications may not be the obvious ones we envision today

Blockchain vs. internet: Internet transformed information flow; blockchain aims to transform value transfer without trusted intermediaries

Timeline: Iansiti and Lakhani (HBR 2017) predicted 10–20 year adoption timeline for enterprise blockchain — similar to internet adoption

Source: Iansiti and Lakhani, HBR 2017



Cryptocurrencies: Market Structure and Institutional Adoption

#	Name	Market Cap	Price	Volume (24h)	Circulating Supply	Change (24h)	Price Graph (7d)
1	 Bitcoin	\$110,373,008,479	\$6,412.06	\$4,402,835,641	17,213,337 BTC	-2.10%	
2	 Ethereum	\$29,514,371,102	\$291.17	\$1,694,294,522	101,365,151 ETH	-2.63%	
3	 XRP	\$11,750,490,940	\$0.298445	\$256,839,252	39,372,399,467 XRP *	1.15%	
4	 Bitcoin Cash	\$9,139,512,523	\$528.41	\$405,914,021	17,296,263 BCH	-0.81%	
5	 EOS	\$4,269,901,440	\$4.71	\$635,275,916	906,245,118 EOS *	-2.24%	
6	 Stellar	\$4,078,148,974	\$0.217249	\$79,689,264	18,771,752,950 XLM *	-4.72%	

Cryptocurrencies: 1854 • Markets: 13025 • Market Cap: \$207,836,652,544 • 24h Vol: \$13,029,131,704 • BTC Dominance: 53.1%

Bitcoin dominance: Bitcoin holds ~50–55% of total crypto market cap (\$1.3 trillion); Ethereum underpins DeFi, NFTs, and tokenized assets

SEC spot ETF approval: Bitcoin spot ETFs approved in January 2024; BlackRock's IBIT accumulated \$17B in its first 8 weeks — fastest ETF launch in history, signaling institutional legitimacy

Source: CoinMarketCap

ICOs: The Billion-Dollar Crowdfunding Experiment

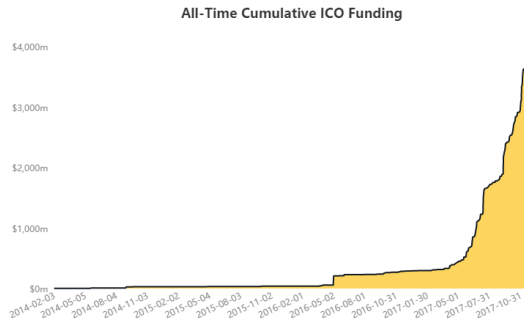
ICO mechanics: Startups issue tokens via smart contract; investors pay in ETH or BTC in exchange for future utility rights or equity-like claims

Scale: ICOs raised \$5B+ in 2017, bypassing investment banks and accredited investor rules to raise capital globally

Key risks: No investor protections, high fraud rate (~80% of 2017 ICOs assessed as scams by academic studies), and extreme regulatory uncertainty

Legacy: Regulatory crackdown (SEC enforcement) killed the ICO model by 2019; replaced by STOs, SAFTs, and now token launches on regulated venues

Source: CoinDesk



How ICOs Work: The Mechanics

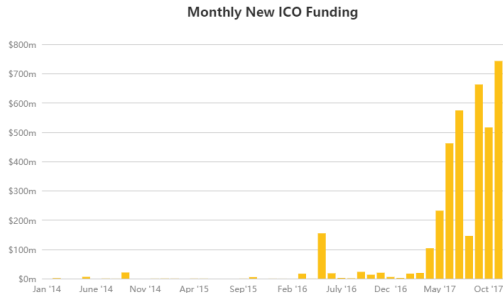
Token issuance: Smart contract automatically issues tokens to investors who send ETH/BTC; transparent, instant, and global

No intermediary: No investment bank, no roadshow, no underwriting fee (typically 3–7% of capital raised saved)

Liquidity: Tokens trade on crypto exchanges immediately post-ICO, providing instant liquidity vs. 7–10 year VC lockup

Investor protection gap: No prospectus, no audited financials, no registration, no recourse if the team disappears with the funds

Source: CoinDesk



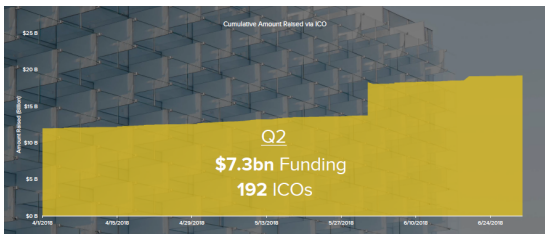
ICO Peak in Q2 2018: Larger Than Most Traditional IPOs

EOS: Raised \$4 billion over 12 months in the largest ICO in history — larger than most traditional IPOs without a single SEC filing

Telegram: Raised \$1.7 billion from accredited investors in a private token offering; SEC forced it to return the funds and pay \$18.5M penalty

Collapse: ICO activity fell 95%+ in late 2018 alongside Bitcoin's crash from \$20K to \$3K, demonstrating tight coupling between crypto sentiment and fundraising

Current landscape: IEOs, STOs, and airdrops replaced ICOs under tighter regulatory frameworks



Global FinTech Investment and Ecosystems

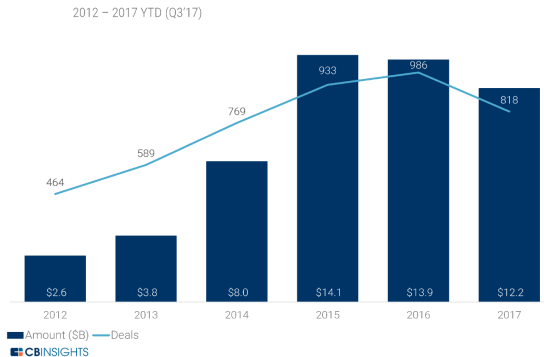
Global FinTech VC Investment: Boom, Peak, and Reset

Peak: Global FinTech VC investment peaked at \$132 billion in 2021 as zero interest rates fueled a venture capital boom

Correction: Investment contracted to \$51B in 2023 as rising rates compressed tech valuations and dried up growth-at-any-cost funding

Winners: Payments, lending, and AI-powered financial services attracted the most capital throughout the cycle

Quality filter: The 2022–23 correction reset valuations toward fundamentals, separating durable businesses from ZIRP-era zombies



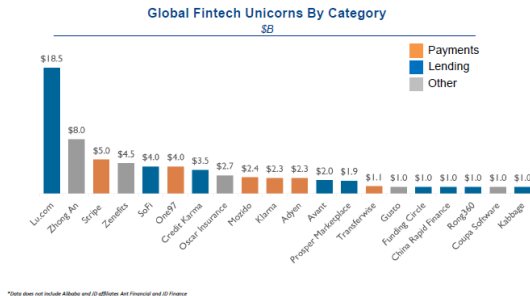
FinTech Unicorns: 300+ Companies Valued Over \$1 Billion

Global count: Over 300 FinTech unicorns globally; collectively valued at \$1.4 trillion at peak

Leaders: Ant Group (\$78B), Stripe (\$65B), and Chime (\$25B) lead private valuations as of 2024

Valuation reset: Post-2022 down rounds exposed unicorns with weak unit economics; Klarna fell from \$46B to \$7B before recovering to \$15B at its 2024 IPO

IPO pipeline: Stripe, Chime, and Revolut are among the most anticipated FinTech IPOs; public markets will be the ultimate test



The Global FinTech Map: Three Dominant Ecosystems



Top 10 US fintech investments

Q3'17

Company	Round	State	Select Investors
 Kabbage	\$250M (Series F)	Georgia	BlueRun Ventures, Reverence Capital Partners, SoftBank Group, Thomvest Ventures, Wildcat Venture Partners
 Upserve	\$151M (Private Equity)	Rhode Island	Vista Equity Partners
 bread	\$126M (Series B)	New York	Bessemer Venture Partners, Menlo Ventures, RRE Ventures
 toast	\$101M (Series C)	Boston	Bessemer Venture Partners, Generation Investment Management, Lead Edge Capital
 blend	\$100M (Series D)	California	8VC, Emergence Capital Partners, Greylock Partners, Lightspeed Venture Partners, Nyca Partners
 coinbase	\$100M (Series D)	California	Battery Ventures, Draper Associates, Greylock Partners, Institutional Venture Partners, Section 32, Spark Capital
Betterment	\$70M (Series E-II)	New York	Bessemer Venture Partners, Francisco Partners, Kinnevik, Menlo Ventures
 Raise	\$60M (Series c)	Illinois	Accel Partners, Bessemer Venture Partners, New Enterprise Associates, PayPal
 PROSPER	\$50M (Growth Equity)	California	FinEX Asia
 highradius	\$50M (Growth Equity)	Texas	Susquehanna Growth Equity

Asia-Pacific: The World's Largest FinTech Market

Investment share: Asia-Pacific accounted for 43% of global FinTech investment in 2018; China alone drove over 60% of the regional total

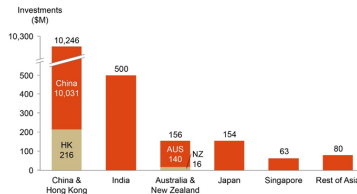
Mobile payments: WeChat Pay and Alipay process more transactions in China than Visa and Mastercard process globally — combined annual volume of \$60+ trillion

Leapfrogging: China bypassed the credit card era entirely — consumers moved directly from cash to mobile payments

Financial inclusion: India's UPI system processed 100 billion transactions in 2023, bringing digital payments to 500 million people

China dominates Asia-Pacific investments

APAC Fintech Financing Activity
(2016, \$M)



Note: Investments value referred to only deals with amount reported by CB Insights
Source: Accenture Research analysis on CB Insights data

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FinTech in China: The Super-App Model

Market dominance: WeChat Pay and Alipay process 97%+ of China's mobile payments, creating unrivaled transaction data assets for AI-based lending and credit scoring

Super-app model: WeChat combines messaging, social media, payments, shopping, ride-hailing, and financial services in one app — 1.3 billion daily active users

Data advantage: More daily behavioral data than any Western financial institution; enables instant AI-based credit decisions

Ant Group scale: Before its blocked IPO, Ant was valued at \$315B — more than JPMorgan Chase



Economist.com

Ant Financial: The World's Largest FinTech Company

Yu'e Bao: Ant Financial's money market fund became the world's largest within 3 years, accumulating \$250B+ in AUM by letting Alipay users earn interest on their balances

Embedded finance: Financial products embedded in super-apps reach scale impossible through standalone offerings — users do not need to “choose” a financial product

Sesame Credit: Ant's proprietary credit scoring system covers 1 billion people without bank credit histories

Regulatory reset: Chinese government blocked Ant's \$37B IPO in 2020 and forced restructuring — regulatory risk for FinTech giants is very real



Economist.com

China's P2P Lending: A Cautionary Tale

Peak scale: China's P2P lending sector reached \$218B in outstanding loans (2017) — the largest in the world by far

Collapse: Regulatory crackdown from 2018 exposed widespread fraud; 6,000+ platforms collapsed; millions of retail investors lost savings

Ezubao fraud: The largest P2P fraud; \$7.6B raised from 900,000 investors in a Ponzi scheme; 21 executives imprisoned

Lesson: Rapid scale without investor protection creates systemic risk; China's P2P collapse is required reading for FinTech risk management

Source: KPMG

Internet attackers and their five key success factors



- "Barbarians from outside"
- Dominated by a few magnets who establish their comprehensive financial eco-systems with different focuses on core businesses (e-Commerce or social network) and target groups
- Numerous small-sized companies with specific positioning in a niche market, especially in wealth management and P2P sectors
- Grow rapidly relying on the huge user base



Singapore: How to Build a FinTech Hub Deliberately

Digital infrastructure: Near-universal broadband, national digital ID (SingPass used by 4M residents), and open banking framework underpin Singapore's digital economy

MAS proactive stance: Monetary Authority of Singapore promotes FinTech through regulatory sandboxes, digital bank licenses (awarded to Grab, Sea, Ant), and a \$225M FinTech investment fund

Talent hub: 1,000+ FinTech firms employ 10,000+ professionals in Singapore; key gateway to Southeast Asia's 680M-person market

Awards: Named Global FinTech Hub of the Year 5 times by Financial Times



Source: Digital Economy Singapore, Presentation by Mohanty S., Monetary Authority of Singapore, 2018

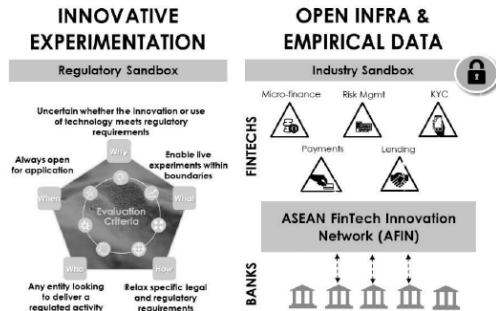
Regulatory Sandboxes: Experimenting Safely with Innovation

MAS sandbox: Launched 2016; allows companies to test products with real customers without full licensing for 6–18 months

Global spread: Over 50 jurisdictions now operate sandbox programs, from UK's FCA Sandbox to HKMA's Fintech Supervisory Sandbox

What it enables: Startups can test AI-based lending, crypto products, and new payment rails with regulatory oversight — reducing legal risk and compliance cost

Success rate: 70%+ of sandbox graduates successfully obtain full licensing and scale their products commercially



Source: Digital Economy Singapore, Presentation by Mohanty S., Monetary Authority of Singapore, 2018

Challenges Facing the FinTech Industry

Incumbent Response

Innovation and competition: JPMorgan spends \$15B/year on technology — more than most FinTech unicorns are worth

Acquisition: Visa's attempted \$5.3B acquisition of Plaid (blocked by DOJ); American Express acquired Kabbage for \$850M

Partnership: Goldman Sachs powers Apple Card; BBVA acquires fintech stakes globally

Technology and Security Risks

Cybersecurity: FinTech firms are high-value targets; Equifax breach exposed 147M records; crypto exchange hacks total \$3B+

Model risk: AI underwriting models can embed bias or fail catastrophically in novel economic conditions

Regulatory Uncertainty

Crypto classification: Are tokens securities (SEC jurisdiction) or commodities (CFTC)? The answer determines the entire regulatory framework

Lending platforms: State-by-state interest rate caps (usury laws) create regulatory arbitrage and compliance complexity

ICOs: SEC enforcement effectively shut down the ICO market; future token launches must navigate securities laws

Key Lesson

The three biggest FinTech risks are regulatory risk, cybersecurity risk, and credit cycle risk — all of which are poorly captured in bull-market performance data.

Research: How Valuable Is FinTech Innovation?

How Valuable Is FinTech Innovation? Chen, Wu, and Yang (2019)

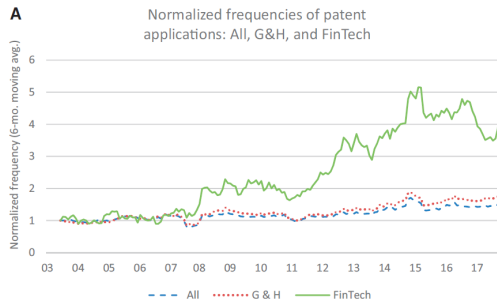
Data: Patent filings from 2003 to 2017 — large-scale evidence on the occurrence and value of FinTech innovation across the entire industry

Key finding 1: FinTech patents grew 42% per year from 2010–2017, far outpacing overall patent growth of ~5%

Key finding 2: Private firms (startups) account for ~70% of FinTech patent filings — innovation is disproportionately driven by new entrants, not incumbents

Key finding 3: Firms that own FinTech patents experience significantly less competitive disruption — patents are a viable incumbent defense strategy

Source: Chen, Wu, and Yang (2019)



FinTech Innovation: Public vs. Private Firms

Startup dominance: Private firms account for ~70% of FinTech patent filings — innovation is disproportionately driven by new entrants

Incumbent focus: Banks and payment networks concentrate patents on payments and cybersecurity; startups patent broadly across lending, blockchain, and AI

Patent value: FinTech patents are associated with large positive announcement returns when granted — markets price the competitive value of innovation

Acquisition target: High patent counts predict future acquisition; FinTech patent portfolios are a key driver of M&A target selection

Source: Chen, Wu, and Yang (2019)

	Number of Patent Applications
FinTech Patents	7,233
Public Firms	3,113
Private Firms	1,287
Individuals	2,833
Financial Services Sector	1,785
Public Financial Services Sector	1,583
Asset Management Companies	10
Banking/Lending Companies	876
Brokerage Companies	25
Insurance Companies	74
Payments Companies	598
Private Financial Services Sector	202
Non-Financial Sector	2,615

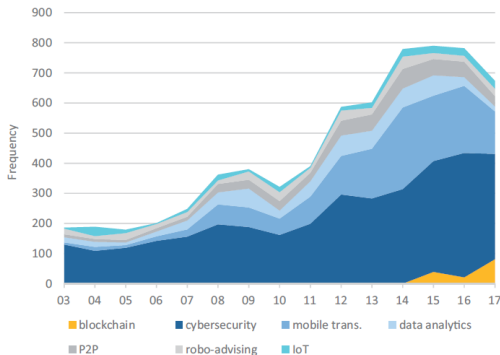
FinTech Patent Filings by Technology Category

Growth rate: FinTech patents grew 42% per year from 2010–2017, far outpacing overall patent growth of ~5%

AI and ML: Fastest growing category — AI-based credit scoring, fraud detection, and trading algorithms dominate recent filings

Blockchain: Second fastest — every major bank has filed blockchain patents despite public skepticism about enterprise adoption

Payments: Largest absolute category — incumbents (Visa, Mastercard) defend existing payment infrastructure through patent thickets



Industry Disruption by FinTech Category

Payments disrupts banking most: The payments category creates the greatest competitive disruption for commercial banks, measured in lost market value

Data analytics targets insurance: AI-based data analytics most disrupts insurance and investment management

Patent defense works: Firms that own FinTech patents experience significantly less competitive disruption — the evidence supports aggressive patent investment as a defensive strategy

Implication for careers: Understanding which FinTech category threatens your sector is the first step to building a defensive strategy

	Industry				
	Banking	Payments	Brokerage	Asset management	Insurance
Most-disruptive technology category	Blockchain [−9,117.5]	IoT [−209.1]	P2P [−591.5]	Blockchain [−535.4]	Robo-advising [−526.7]
2nd-most-disruptive technology category	Cybersecurity [−368.2]	Blockchain [−74.9]	Robo-advising [−334.6]	IoT [−253.2]	P2P [−358.6]
Diff. in medians (top-two disruptive categories vs. others)	−4,744.0	−224.5	−427.7	−270.6	−418.4
p-value, permutation test for diff. in medians	.021	<.001	<.001	<.001	.028

Visa's FinTech Acquisition Strategy

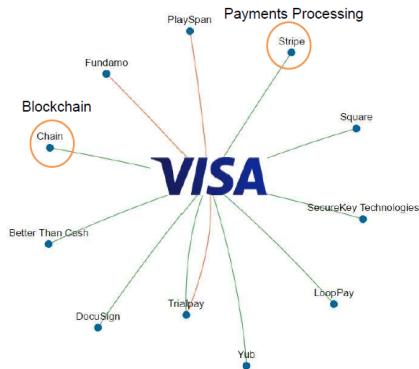
Acquisition spree: Visa has acquired 30+ FinTech companies since 2015, including CyberSource (\$2B), Earthport (\$247M), and Plaid (\$5.3B, blocked by DOJ)

Strategic logic: Each acquisition extends Visa's payment infrastructure into new data layers, geographies, or customer segments

DOJ block on Plaid: The antitrust challenge signaled growing regulatory scrutiny of financial platform consolidation — Plaid would have given Visa access to 11,000 banks' account data

After Plaid: Visa acquired Currencycloud (\$963M) and Tink (\$1.8B) to rebuild its open banking capability organically

Source: CB Insights



Mastercard's FinTech Acquisitions: Targeting Real-Time Payments

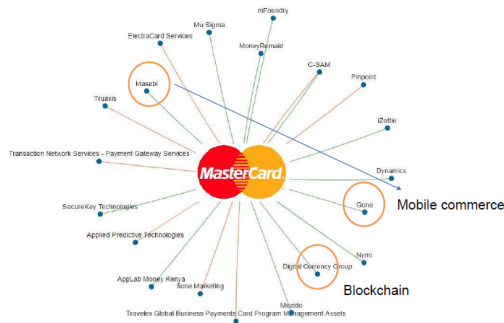
Vocalink: Acquired for \$920M in 2016; operates the UK's Faster Payments infrastructure and the U.S. ATM networks used by 10,000+ banks

Nets: Danish payment company providing real-time payment infrastructure across Northern Europe

Finicity: Open banking data platform; provides bank-permissioned financial data access to lenders and apps

Strategic theme: Both Visa and Mastercard are hedging against account-to-account payment systems (like FedNow) that could bypass card rails entirely and eliminate interchange revenue

Source: CB Insights



Research: FinTech Mergers and Value Creation

FinTech Mergers and Value Creation: Wang and Zhang (2022)

Research question: Why do traditional firms acquire FinTech companies, and does this create value for their shareholders?

Data: FinTech M&A deals from 2000–2018 involving publicly listed acquirers

Setting: Traditional financial and non-financial firms acquiring FinTech targets

Key findings:

FinTech firms are increasingly sought-after acquisition targets

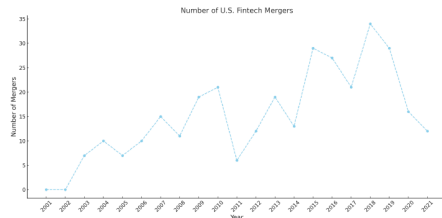
Acquirers face FinTech disruption *before* the deal closes

Mergers generate significant positive announcement returns

Non-acquiring industry peers also benefit — positive spillovers

Figure 1: Time Series Distribution of FinTech Mergers

This figure illustrates the time series distribution of FinTech mergers from 2003 to 2021. FinTech mergers are defined as deals whose targets are FinTech firms.



Finding 1: FinTech Firms Are Highly Sought-After Acquisition Targets

	(1)	(2)	(3)	(4)
Dependent Variable	<i>Prob(M&A Target)</i>			
<i>FinTech</i>	0.0055*** (0.002)	0.0038** (0.002)	0.0057*** (0.002)	0.0048*** (0.002)
<i>Size</i>	0.0015*** (0.000)	0.0014*** (0.000)	0.0013*** (0.000)	0.0013 (0.001)
<i>ROA</i>	0.0009*** (0.000)	0.0010*** (0.000)	0.0010*** (0.000)	-0.0001 (0.000)
<i>Leverage</i>	-0.0049 (0.003)	-0.0062** (0.003)	-0.0036 (0.003)	-0.0076** (0.003)
<i>Market-to-book</i>	-0.0000* (0.000)	-0.0000 (0.000)	0.0000 (0.000)	-0.0000 (0.000)
<i>R&D</i>	0.0104*** (0.003)	0.0081*** (0.003)	0.0078*** (0.002)	0.0082*** (0.003)
Observations	81,429	75,796	75,225	80,222
R-squared	0.003	0.009	0.052	0.201
Industry FE	No	Yes	No	No
Firm FE	No	No	No	Yes
Industry×Year FE	No	No	Yes	No
Year FE	Yes	Yes	No	Yes

Key Takeaway

FinTech M&A volume grew 300%+ from 2013 to 2018. Both financial and non-financial incumbents (retailers, telecoms, BigTech) are acquiring FinTech startups, reflecting broad cross-industry disruption potential. The pattern mirrors how incumbents responded to the internet in the late 1990s — buying what they could not build organically.

Finding 2: Acquirers Face FinTech Disruption Before the Merger

	(1)	(2)	(3)
Variables	<i>Combined FinTech Firm</i>		
Sample	Full	<i>Diversify = 1</i>	<i>Financial Acquirer = 0</i>
<i>FinTech Merger</i>	0.1344*** (0.045)	0.1677*** (0.052)	0.1591*** (0.054)
<i>FinTech Acquirer</i>	0.4442*** (0.042)	0.4765*** (0.049)	0.4451*** (0.040)

Key Takeaway

Acquiring firms show declining market share, revenue growth, and profitability in the 3–5 years *before* completing a FinTech acquisition. FinTech M&As are primarily **defensive** strategic responses to competitive pressure, not opportunistic growth moves. Disruption is the catalyst; acquisition is the response.

Finding 3: Significant Shareholder Value Creation

Announcement CARs are significantly *positive* for acquirers

Markets reward FinTech acquisitions more than comparable non-FinTech deals

Value creation is larger when:

Acquirer faces greater disruption threat

Target holds more innovative patents

Deal involves technology integration

Variables	(1) Combined CAR	(2) Target CAR	(3) Acquirer CAR	(4) Combined Dollar Gains	(5) Target Dollar Gains	(6) Acquirer Dollar Gains
<i>FinTech Merger</i>	0.0077* (0.004)	0.0308* (0.016)	0.0021 (0.004)	0.2775*** (0.074)	0.2360*** (0.073)	0.0278 (0.030)
<i>Relative size</i>	0.0071 (0.007)	-0.0270 (0.019)	0.0012 (0.007)	-0.1928** (0.082)	-0.2098** (0.073)	0.0173 (0.046)
<i>Hostile</i>	0.0333** (0.015)	0.1033** (0.044)	0.0200 (0.017)	0.6463* (0.309)	0.5025* (0.256)	0.1428 (0.124)
<i>Diversify</i>	-0.0002 (0.004)	0.0040 (0.008)	-0.0041 (0.005)	0.0377 (0.064)	0.0705 (0.044)	-0.0339 (0.039)
<i>Acquirer size</i>	-0.0063*** (0.002)	0.0371*** (0.009)	0.0042* (0.002)	0.2187*** (0.044)	0.1990*** (0.044)	0.0244 (0.014)
<i>Target size</i>	0.0013 (0.001)	-0.0548*** (0.007)	-0.0071*** (0.002)	-0.1691*** (0.037)	-0.1014*** (0.033)	-0.0650*** (0.016)
<i>Acquirer R&D</i>	-0.1245*** (0.041)	-0.2216 (0.146)	-0.1318* (0.066)	-1.4415* (0.750)	-0.6501 (0.595)	-0.8185** (0.366)
<i>Target R&D</i>	-0.0141 (0.022)	0.1974 (0.131)	0.0067 (0.037)	0.7256 (0.628)	0.8852 (0.575)	-0.0349 (0.194)

Key Takeaway

Investors interpret FinTech acquisitions as credible signals of strategic repositioning — pricing in the firm's forward-looking ability to compete in an increasingly digital financial services landscape.

Finding 4: Positive Spillover Effects on Industry Peers

Non-acquiring peers also earn positive abnormal returns when a rival completes a FinTech deal

Suggests *industry-wide* value, not only acquirer-specific gains

Mechanism: the deal signals FinTech viability, reducing uncertainty sector-wide

Key Takeaway

FinTech acquisitions create positive externalities for the financial industry. Each deal reduces uncertainty about the value of FinTech integration and legitimizes the technology for all industry participants.

	(1)	(2)	(3)
Variables	$\#Mergers_{t+1}$	$\#Mergers_{t+2}$	$\#Mergers_{t+3}$
<i>FinTech Merger</i>	0.0954*** (0.029)	0.1593*** (0.049)	0.2318*** (0.067)
<i>Relative size</i>	-0.0397 (0.034)	-0.0764 (0.044)	-0.0802 (0.061)
<i>Hostile</i>	0.0102 (0.143)	-0.0869 (0.214)	-0.0576 (0.276)
<i>Diversify</i>	-0.2301*** (0.045)	-0.3831*** (0.051)	-0.5027*** (0.065)
<i>Acquirer size</i>	0.0216 (0.014)	0.0275 (0.018)	0.0459** (0.021)
<i>Target size</i>	-0.0009 (0.011)	-0.0108 (0.014)	-0.0155 (0.019)
<i>Acquirer R&D</i>	1.3935*** (0.425)	1.6896** (0.588)	2.6524*** (0.693)
<i>Target R&D</i>	0.7467***	1.1203***	1.6158***

Incumbents' Response to Robo-Advisory Competition

Trigger: After Betterment and Wealthfront proved the robo-advisor model, Schwab, Vanguard, Goldman, and JPMorgan launched competing services within 2–3 years

Incumbent advantages: Existing client bases, brand trust, and regulatory standing let incumbents dominate AUM rapidly once they committed

Schwab result: Intelligent Portfolios accumulated \$60B+ in AUM by 2020 — dwarfing Betterment and Wealthfront's combined total

Lesson: Decisive incumbents with scale can recapture ground when they fully commit to a response, not a half-measure

Source: Investment News



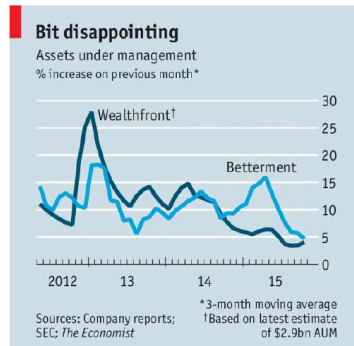
Robo-Advisory AUM Growth: Slowing as Incumbents Dominate

Growth slowdown: Pure-play robo-advisor growth is slowing as bank-owned platforms intensify competition and compress margins across the industry

Consolidation phase: FinTech industries follow a predictable pattern: rapid entry, then consolidation as network effects and scale advantages concentrate market share

Profitability pressure: Most pure-play robo-advisors have not reached profitability; low fee rates require massive AUM to cover fixed technology and compliance costs

Exit paths: Betterment remains independent; Wealthfront was acquired by UBS for \$1.4B; most smaller robo-advisors have been acquired or shut down



Economist.com

Career Opportunities in FinTech

Why FinTech Will Affect Every Finance Career

Roles Being Created

FinTech analyst: Evaluating technology adoption and disruption risk (IBD, consulting)

Product manager: Building financial apps; requires coding, design, and finance knowledge — rare and highly paid

Data scientist: ML-based credit scoring, fraud detection, algorithmic trading; Python/R skills essential

Blockchain developer: Smart contracts, DeFi protocols; Solidity and Web3 command significant premiums

Skills That Matter

Coding: Python for data analysis; SQL for database queries; even non-technical roles benefit from reading code

Financial knowledge: FinTech without finance is just technology; credit risk, valuation, and regulation remain essential

AI literacy: Understanding ML model outputs, limitations, and risks — required in every senior finance role by 2030

Key Lesson

The most valuable FinTech professionals combine deep financial knowledge with technical skills. Neither alone is sufficient in 2025.

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